

# SIH26185 PROJECT SUMMARY

Lightweight Dual-Band Helmet-Integrated Tactical Antenna (D-HITA)

Problem: Helmet mounted conformal antenna for tactical communications in urban CQB environments

Organization: Ministry of Home Affairs · National Security Guard (NSG), Police II Division

Category: Hardware · Theme: Robotics and Drones

Coverage: Project work and decisions through September 2026

**Purpose: Single handover-ready record of requirements, research, engineering decisions, work completed, current status and next actions.**

Confirmed source information is separated from engineering assumptions and future work. Unresolved items are marked [TO BE CLARIFIED].

# 1. PROJECT OVERVIEW

## 1.1 Problem Statement

SIH26185 addresses tactical communications for NSG commandos operating in restrictive urban CQB environments such as closed rooms, basements, narrow corridors and stairwells. The PS identifies vest-mounted handheld radios with rigid, protruding omnidirectional whip antennas as causing snagging/obstruction and RF limitations. In reinforced-concrete or steel/glass buildings, low antenna positioning can cause attenuation/fading; omnidirectional radiation can also increase vulnerability to eavesdropping or directional tracking. The PS therefore calls for moving the antenna to the helmet while maintaining a low-profile form and ballistic integrity.

The expected solution is a ruggedized, zero-profile wearable antenna using lightweight, ultra-thin, flexible microstrip patch elements integrated with tactical ballistic helmets; support UHF and L-band operation; use underside RF shielding to isolate the head and direct radiation upward/outward; and route a ruggedized coaxial interface cleanly along the helmet to existing NSG handheld radios and body-worn camera systems.

Source: SIH26185 problem statement, reviewed 10 September 2026. [cite]turn0search3[

## 1.2 Exact PS requirements

| Requirement              | Confirmed PS meaning                                                        |
|--------------------------|-----------------------------------------------------------------------------|
| Helmet integration       | Integrate seamlessly into/on tactical ballistic helmets.                    |
| Low / zero profile       | Low-profile; ruggedized zero-profile/minimal protrusion.                    |
| Flexible / conformal     | Flexible elements that conform to helmet geometry.                          |
| Lightweight / ultra-thin | Lightweight, ultra-thin microstrip patch elements.                          |
| Conformal array          | Expected solution is a conformal antenna array.                             |
| UHF                      | Must support UHF; exact numerical endpoints are not stated.                 |
| L-band                   | Must support L-band; exact numerical endpoints are not stated.              |
| RF shielding             | Underside shielding to isolate the head from radiation.                     |
| Radiation direction      | Optimized radiation pattern upward and outward.                             |
| High gain                | Problem description calls for high gain; numeric target not stated.         |
| Ruggedized construction  | Ruggedized wearable antenna system.                                         |
| Coax interface           | Ruggedized coax routed cleanly along helmet.                                |
| Existing equipment       | Direct connection to existing handheld radios and body-worn camera systems. |
| Live video context       | UHF/L-band intended to support radio and camera/live-video equipment.       |
| Protection               | Must not degrade ballistic or impact protection ratings.                    |

**Important: The PS does not state 300 MHz-3 GHz, nor exact gain, bandwidth, range, substrate, shield material, connector, dimensions or weight. These remain engineering targets to be derived. [TO BE CLARIFIED]**

## 1.3 Project title and objectives

- Preferred title: Lightweight Dual-Band Helmet-Integrated Tactical Antenna
- Acronym: D-HITA — Dual-band Helmet-Integrated Tactical Antenna.
- Develop a low-profile, flexible, conformal antenna architecture for ballistic-helmet integration.
- Target UHF + L-band while preserving helmet protective function.
- Use shielding and geometry to reduce inward radiation and favor upward/outward coverage.
- Connect to existing equipment through ruggedized coax.
- Validate through EM simulation, fabrication and RF measurement.

## 1.4 Team composition

A final named team roster is not present in the available record. Proposed roles: RF/EM, simulation/optimization, mechanical/CAD, fabrication/electronics, testing, research/documentation, system lead. [TO BE CLARIFIED]

## 2. INITIAL ANALYSIS & REQUIREMENTS

### 2.1 Initial analysis

- Treat the problem as an RF + mechanical integration problem rather than a software project.
- Use the helmet as the highest practical mounting point while avoiding a protruding whip.
- Use a passive antenna baseline; no dedicated helmet battery, MCU or power amplifier is required in the baseline concept.
- Balance low profile, flexibility, conformality, UHF physical size, L-band operation, shielding, gain, bandwidth and helmet area.
- Do not claim novelty before comparing against prior art; reproduce a credible baseline first and then engineer measurable improvements.

### 2.2 Working technical requirements

| Area      | Working requirement                                             | Status                                          |
|-----------|-----------------------------------------------------------------|-------------------------------------------------|
| RF        | UHF + L-band                                                    | PS confirmed; endpoints [TO BE CLARIFIED]       |
| Interface | 50 $\Omega$ RF path                                             | Engineering target                              |
| Form      | Flexible, conformal patch/array                                 | PS-derived                                      |
| Profile   | Minimal protrusion                                              | PS-derived; thickness [TO BE CLARIFIED]         |
| Shield    | Conductive underside layer                                      | PS-derived; geometry/material [TO BE CLARIFIED] |
| Radiation | Upward/outward, high gain                                       | PS-derived; numeric targets [TO BE CLARIFIED]   |
| Feed      | Ruggedized coax                                                 | PS-derived; connector [TO BE CLARIFIED]         |
| Testing   | S11/VSWR, bandwidth, gain, efficiency, pattern; SAR if feasible | Planned                                         |

### 2.3 Constraints

- Ballistic and impact protection cannot be degraded.
- Minimal protruding hardware to reduce snagging.
- Helmet curvature and head loading must be included in later simulation.
- Shielding is a trade-off: it can reduce head-directed fields but can also detune/reduce efficiency.
- UHF wavelengths make compact antennas difficult; loading, slots, meanders, PIFA/folded or multi-resonant approaches may be needed.
- Baseline integration avoids drilling/cutting the protective shell; mounting method remains [TO BE CLARIFIED].

### 2.4 KPIs

| KPI                 | Evidence                                                       |
|---------------------|----------------------------------------------------------------|
| Resonance           | Target-band resonant frequency                                 |
| S11 / VSWR          | Measured and simulated matching; threshold [TO BE CLARIFIED]   |
| Bandwidth           | Frequency span meeting matching criterion                      |
| Gain / efficiency   | Measured/simulated values; targets [TO BE CLARIFIED]           |
| Radiation           | Upward/outward pattern and reduced head-directed radiation     |
| Size/profile/weight | Helmet footprint, thickness and mass; limits [TO BE CLARIFIED] |
| Ruggedness          | Mechanical/environmental validation plan [TO BE CLARIFIED]     |

## 3. WORK COMPLETED (CHRONOLOGICAL ORDER)

### 3.1 Problem analysis and requirement freeze

Early September 2026: SIH26185 was re-read and converted into a requirement checklist. A key correction was made: the official PS gives only UHF and L-band, not a 300 MHz–3 GHz range.

### 3.2 SIH presentation and project documentation

- A six-slide SIH-style structure was developed: title, proposed solution, technical approach, feasibility/viability, impact/benefits, research/references.
- An internal project-bible approach was established covering architecture, RF basics, components, simulation, risks, roles, Q&A; and checklists.
- The uploaded Recommended Presentation Flow emphasizes problem → solution → technical approach → architecture → technology stack → innovation → feasibility → impact → references, with visual evidence over heavy text. `filecite turn8file0`

### 3.3 Research completed

| Reference                                                                                                                                                      | Role in project                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Herold, Griffiths, Fung et al., IEEE MILCOM 2007, “Lightweight, High-Bandwidth, Ultra-Wideband Antenna for Ballistic Helmets.” DOI 10.1109/MILCOM.2007.4414109 | Direct baseline helmet antenna system for ballistic helmets. |
| Lebaric & Tan, IEEE MILCOM 2000, “Ultra-wideband RF helmet antenna.” DOI 10.1109/MILCOM.2000.905081                                                            | Early baseline helmet antenna.                               |
| Li et al., 2023, “Research Progress on Helmet Antenna in Individual Soldier Communication System.” DOI 10.1199/JET220613                                       | Helmet communication system background.                      |
| Nepa & Rogier, IEEE AP Magazine 2015. DOI 10.1109/MAP.2015.2472377                                                                                             | Wearable/conformal antenna background.                       |
| Tang et al., IEEE TAP 2025, “A Helmet-Based Hemispherical-Conformal Wearable Antenna.” DOI 10.1109/TAP.2025.2560101                                            | Wearable helmet antenna.                                     |

Research conclusion: the five papers already read are enough to move into a structured comparison and baseline selection. More papers should be searched only for missing evidence, such as UHF helmet antennas, dual-band UHF/L-band designs, shielding/head loading, and patents.

### 3.4 NSG procurement/equipment investigation

NSG's public tender archive lists Digital UHF Hand Held Radio Sets (20 W and 4 W), a Digital UHF Repeater Set (40 W), and a Wireless Body Worn Video System in 2026. This supports the operational relevance of UHF radio and body-worn video. `cite turn0search0`

The accessible evidence did not establish the exact NSG radio frequency range, exact awarded radio model, or exact L-band frequency used by the relevant video equipment. Therefore 400–527 MHz, investigated earlier as a possible Motorola R7 UHF range, must not be treated as the confirmed NSG requirement. [TO BE CLARIFIED]

### 3.5 Architecture and component stack

Baseline stack, outward to inward: protective film → copper antenna → flexible substrate → copper ground/shield → mounting adhesive → helmet shell → head.

Candidate physical components: ballistic-style helmet/shell; flexible dielectric; conductive antenna; conductive shield; dielectric spacer/foam; 50 Ω coax; RF connector; strain relief/cable clips; non-metal adhesive; protective outer layer; optional matching network; flex-PCB feed.

### 3.6 Toolchain and setup

| Tool                                | Purpose                               |
|-------------------------------------|---------------------------------------|
| Ansys HFSS Student                  | Primary EM simulation                 |
| CST Studio Suite Learning Edition   | Alternative EM simulation             |
| KiCad                               | PCB/flex-PCB and RF feed layout       |
| SolidWorks / FreeCAD                | Helmet CAD/mechanical integration     |
| Abaqus 2026                         | Optional mechanical FEA               |
| Python / NumPy / SciPy / Matplotlib | Analysis/optimization                 |
| Figma                               | System diagrams/concept visualization |
| Keysight ADS                        | Optional RF/matching work             |

Abaqus 2026 was installed successfully after resolving its Java requirement. Oracle JDK 21.0.12 was used as the supported Java option; Android Studio's bundled JBR was left untouched. A Dassault Environment Editor was accidentally opened while searching for CST and was identified as unrelated; no environment variables were to be modified.

### 3.7 Prototype methodology defined

Prototype ladder: P0 simulation → P1 flat antenna → P2 curved antenna → P3 helmet-mounted → P4 helmet + head → P5 tested prototype → P6 final demonstrator.

Core workflow: requirements → research → candidate architectures → EM simulation → optimization → helmet/head/shield model → fabrication → VNA/radiation testing → iteration → validation.

## 4. KEY FINDINGS & RESULTS

### 4.1 Findings

- Helmet mounting is intended to raise the antenna relative to the vest-mounted radio and improve the practical propagation geometry.
- The main challenge is the simultaneous satisfaction of UHF + L-band + low profile + flexibility + curvature + shielding + ruggedness + ballistic protection.
- A passive antenna is the correct baseline architecture; no separate helmet battery/electronics are required unless the architecture later becomes active/reconfigurable.
- Shielding should be evaluated as an electromagnetic trade-off, not described as producing zero radiation toward the head.
- Curvature, head loading, cable routing, connector placement and manufacturing tolerance must be included in final validation.
- The strongest project strategy is to reproduce a literature-backed baseline and then show measurable modifications against SIH26185 requirements.

### 4.2 Results not yet available

- No validated numerical UHF/L-band endpoints.
- No HFSS/CST simulation results.
- No S11/VSWR/gain/efficiency/radiation measurements.
- No physical antenna prototype.
- No helmet-mounted RF test.
- No head-phantom/SAR result.
- No final topology or defensible novelty claim.

### 4.3 Requirement comparison

| Requirement                    | Current state                                                |
|--------------------------------|--------------------------------------------------------------|
| Helmet integration             | Architecture defined; not physically built.                  |
| Low profile/flexible/conformal | Design direction defined; dimensions/materials not final.    |
| UHF + L-band                   | Required by PS; endpoints [TO BE CLARIFIED].                 |
| Shielding                      | Layer concept defined; material/geometry not final.          |
| Upward/outward radiation       | Objective defined; no proof yet.                             |
| High gain                      | Required; numeric target/evidence [TO BE CLARIFIED].         |
| Rugged coax                    | Architecture defined; connector selection [TO BE CLARIFIED]. |
| Protection                     | Constraint recognized; no physical validation yet.           |
| Radio/camera compatibility     | Concept defined; exact equipment [TO BE CLARIFIED].          |

### 4.4 Challenges and resolutions

| Challenge                      | Handling                                                                              |
|--------------------------------|---------------------------------------------------------------------------------------|
| RF unfamiliarity               | Decompose into RF fundamentals, literature, simulation, CAD, fabrication and testing. |
| Unclear numerical frequencies  | Stop using 300 MHz–3 GHz as an official PS requirement; obtain equipment evidence.    |
| Too much literature collection | Five papers are enough to move to comparison; search only for gaps.                   |
| Premature novelty claim        | Use a validated baseline and measure modifications.                                   |
| Software setup confusion       | Abaqus/JDK issue resolved; Dassault Environment Editor identified as unrelated.       |

## 5. CURRENT STATUS

### 5.1 Working well

- Problem understanding and requirement decomposition are substantially complete.

- Five relevant papers have been identified and read.
- NSG procurement evidence confirms current UHF handheld-radio and body-worn-video procurement. [\[cite\turn0search0\]](#)
- Simulation, CAD, PCB and analysis tools have been selected and setup has begun.
- Project management is structured around requirements, tasks, research, design, experiments, evidence, decisions, risks and documentation.

## 5.2 What still needs improvement

- Lock exact UHF and L-band frequencies.
- Build the research comparison matrix.
- Identify 3–5 defensible research gaps.
- Select 1–2 baseline architectures.
- Reproduce the baseline in HFSS/CST.
- Define numerical pass/fail thresholds.
- Build helmet CAD and simulate curvature/head/shield.
- Fabricate and VNA-test V1.

## 5.3 Current blockers

- Exact UHF frequency range [TO BE CLARIFIED].
- Exact L-band frequency range [TO BE CLARIFIED].
- Exact NSG radio model(s) [TO BE CLARIFIED].
- Exact body-camera/video RF link [TO BE CLARIFIED].
- Meaning of 'svl equipment' [TO BE CLARIFIED].
- Gain, bandwidth, VSWR/S11, size, thickness and weight limits [TO BE CLARIFIED].
- Helmet model/shell material and approved mounting method [TO BE CLARIFIED].
- Ruggedness/environmental test standard [TO BE CLARIFIED].
- Final team roster [TO BE CLARIFIED].

# 6. NEXT WORKFLOW & ACTION ITEMS

## 6.1 Immediate priorities

P0 – Frequency evidence — Obtain credible UHF/L-band target ranges from equipment specifications, manuals, tender documents or authoritative sources.

P1 – Research matrix — Compare the five papers on frequency, topology, dimensions, substrate, thickness, gain, bandwidth, efficiency, conformality, helmet integration, shielding, testing and limitations.

P2 – Gap analysis — Identify 3–5 gaps created by the combined SIH constraints.

P3 – Baseline selection — Select one primary and one backup antenna architecture.

P4 – Reproduce — Build the published baseline in HFSS/CST and compare to reported behavior.

P5 – Modify — Tune geometry/material/loading to the real target bands.

P6 – Helmet model — Add curvature, shell, shield and head loading.

P7 – Optimize — Run parameter sweeps for resonance, matching, bandwidth, gain, efficiency and pattern.

P8 – Fabricate V1 — Use flexible copper/dielectric or flex PCB.

P9 – VNA — Measure S11, VSWR, resonance and bandwidth.

P10 – Radiation — Measure gain/pattern/efficiency in a suitable lab if available.

P11 – Safety/integration — Evaluate cable routing, mechanical attachment, head exposure and protection constraints.

## 6.2 Timeline

| Window   | Planned work                                                |
|----------|-------------------------------------------------------------|
| Days 1-2 | Frequency evidence + research matrix + requirements freeze. |
| Days 3-4 | Baseline selection + HFSS/CST reproduction.                 |
| Days 5-7 | Dual-band modifications + parameter sweeps.                 |
| Week 2   | Curved helmet + shield + head model; select V1.             |
| Week 3   | Fabrication + VNA.                                          |
| Week 4   | Radiation/efficiency + V2 optimization + documentation.     |

Long-term goal: final dual-band helmet-integrated demonstrator with simulation-vs-measurement evidence and complete requirement traceability.

## 7. TECHNICAL SPECIFICATIONS

| Parameter  | Current specification                                                                            |
|------------|--------------------------------------------------------------------------------------------------|
| Antenna    | Flexible, low-profile, conformal microstrip patch-based array; exact topology [TO BE CLARIFIED]. |
| Bands      | UHF + L-band; numerical endpoints [TO BE CLARIFIED].                                             |
| Feed       | 50 $\Omega$ coax; exact connector [TO BE CLARIFIED].                                             |
| Shield     | Conductive underside layer for head isolation and radiation shaping.                             |
| Radiation  | Upward/outward; high gain; numerical targets [TO BE CLARIFIED].                                  |
| Profile    | Minimal protrusion; thickness [TO BE CLARIFIED].                                                 |
| Weight     | Lightweight; maximum [TO BE CLARIFIED].                                                          |
| Substrate  | Flexible dielectric; material/thickness [TO BE CLARIFIED].                                       |
| Helmet     | Tactical ballistic helmet; exact model/material [TO BE CLARIFIED].                               |
| Power      | Passive antenna baseline; no dedicated helmet battery.                                           |
| Validation | HFSS/CST + VNA + radiation/efficiency; SAR/head phantom if feasible.                             |

### Candidate architectures

- Flexible dual-band microstrip patch
- Conformal PIFA
- Conformal slot
- Dual-resonant patch
- Multi-element conformal array
- Antenna + integrated shielding layer

### Design approach

Use an existing validated antenna as the baseline. A published  $\sim 3$  GHz conformal design can be studied as an architectural starting point, but moving from  $\sim 3$  GHz to UHF is not a simple scaling exercise: wavelength grows substantially and the helmet footprint becomes dominant. Compact loading, slots, meanders, shorting/PIFA structures, folded geometries and multi-resonant approaches may therefore be investigated.

## 8. REFERENCES & RESOURCES

### 8.1 Official / operational sources

- SIH26185 problem statement, Ministry of Home Affairs / NSG. [cite\turn0search3]
- NSG tender archive, including Digital UHF Hand Held Radio Sets, UHF repeater and Wireless Body Worn Video System. [cite\turn0search0]

### 8.2 Papers

- Herold, Griffiths, Fung et al., IEEE MILCOM 2007, "Lightweight, High-Bandwidth Conformal Antenna System for Ballistic Helmets." DOI 10.1109/MILCOM.2007.4455055.
- Lebaric & Tan, IEEE MILCOM 2000, "Ultra-wideband RF helmet antenna." DOI 10.1109/MILCOM.2000.905081.
- Li et al., 2023, "Research Progress on Helmet Antenna in Individual Soldier Communication System." DOI 10.11999/JEIT220613.
- Nepa & Rogier, IEEE Antennas & Propagation Magazine, 2015. DOI 10.1109/MAP.2015.2472374.
- Tang et al., IEEE TAP 2025, "A Helmet-Based Hemispherical-Conformal Wearable Array Antenna ..." DOI 10.1109/TAP.2025.3560181. Full title [TO BE RE-VERIFIED].

### 8.3 Tools

- Ansys HFSS Student
- CST Studio Suite Learning Edition
- KiCad
- SolidWorks / FreeCAD
- Abaqus 2026
- Python / NumPy / SciPy / Matplotlib
- Figma
- Keysight ADS (optional)

### 8.4 Documentation resources

The uploaded Recommended Presentation Flow emphasizes visual problem/solution framing, technical approach, architecture, technology stack, feasibility, impact, references and evidence. [filecite\turn8file0]  
The uploaded SIH demo presentation was used as a reference for six-slide structure and research/reference presentation. [filecite\turn8file4\turn8file7]

### 8.5 Traceability structure

Requirement → Task → Design → Experiment/Test → Evidence → Decision → Documentation

## APPENDIX - CRITICAL OPEN ITEMS

- Exact UHF range
- Exact L-band range
- Exact radio model(s)
- Exact camera/video RF link
- Meaning of 'svl equipment'
- Gain target
- Bandwidth criterion
- S11/VSWR threshold
- Maximum thickness
- Maximum mass
- Helmet model/material
- Shield material
- Ruggedness/environmental standard

- Ballistic/impact validation method
- Final team roster
- Final antenna topology/fabrication route

END OF PROJECT SUMMARY